Package 'betaSandwich'

November 5, 2026

Title Robust Confidence Intervals for Standardized Regression Coefficients

Version 1.0.9.9000

Description Generates robust confidence intervals for standardized regression coefficients using heteroskedasticity-consistent standard errors for models fitted by lm() as described in Dudgeon (2017) <doi:10.1007/s11336-017-9563-z>. The package can also be used to generate confidence intervals for R-squared, adjusted R-squared, and differences of standardized regression coefficients. A description of the package and code examples are presented in Pesigan, Sun, and Cheung (2023) <doi:10.1080/00273171.2023.2201277>.

URL https://github.com/jeksterslab/betaSandwich, https://jeksterslab.github.io/betaSandwich/

BugReports https://github.com/jeksterslab/betaSandwich/issues

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Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 3.5.0)

Suggests knitr, rmarkdown, testthat, betaDelta, betaMC, betaNB

Config/roxygen2/version 8.1.1.9000

NeedsCompilation no

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`BetaADF` *Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach*

Description

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach

Usage

```
BetaADF(object, alpha = c(0.05, 0.01, 0.001), std_y = TRUE, std_x = NULL)
```

Arguments

`object`	Object of class `lm`.
`alpha`	Numeric vector. Significance level α.
`std_y`	Logical. Standardize y. With `std_y = FALSE`, y is left in its original units ($b_j\sigma_{x_j}$).
`std_x`	Character vector or `NULL`. Names of the regressors to standardize. `NULL` (default) standardizes all regressors; `character(0)` standardizes none (b_j/σ_y). Leaving a dummy variable out keeps its 0-1 contrast. The delta method accounts for the sampling variability of the standard deviations.

Details

Note that while the calculation in `BetaADF()` is different from `betaDelta::BetaDelta()` with `type = "adf"`, the results are numerically equivalent. `BetaADF()` is appropriate when sample sizes are moderate to large (`n > 250`). `BetaHC()` is recommended in most situations.

Value

Returns an object of class `betasandwich` which is a list with the following elements:

call Function call.

args Function arguments.

lm_process Processed `lm` object.

gamma_n Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

gamma_hc Asymptotic covariance matrix HC correction.

gamma Asymptotic covariance matrix of the sample covariance matrix.

acov Asymptotic covariance matrix of the standardized slopes.

vcov Sampling covariance matrix of the standardized slopes.

est Vector of standardized slopes.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Browne, M. W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, *37*(1), 62–83. doi:10.1111/j.20448317.1984.tb00789.x

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

See Also

Other Beta Sandwich Functions: `BetaHC()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaADF(object)
# Methods ------------------------------------------------------
print(std)
summary(std)
```

```
coef(std)
vcov(std)
confint(std, level = 0.95)
```

`BetaHC` *Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach*

Description

Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach

Usage

```
BetaHC(
  object,
  type = "hc3",
  alpha = c(0.05, 0.01, 0.001),
  std_y = TRUE,
  std_x = NULL,
  g1 = 1,
  g2 = 1.5,
  k = 0.7
)
```

Arguments

`object`	Object of class `lm`.
`type`	Character string. Correction type. Possible values are `"hc0"`, `"hc1"`, `"hc2"`, `"hc3"`, `"hc4"`, `"hc4m"`, and `"hc5"`.
`alpha`	Numeric vector. Significance level α.
`std_y`	Logical. Standardize y. With `std_y = FALSE`, y is left in its original units ($b_j \sigma_{x_j}$).
`std_x`	Character vector or `NULL`. Names of the regressors to standardize. `NULL` (default) standardizes all regressors; `character(0)` standardizes none (b_j / σ_y). Leaving a dummy variable out keeps its 0-1 contrast. The delta method accounts for the sampling variability of the standard deviations.
`g1`	Numeric. `g1` value for `type = "hc4m"`.
`g2`	Numeric. `g2` value for `type = "hc4m"`.
`k`	Numeric. Constant `k` for `type = "hc5"` $0 \leq k \leq 1$.

Value

Returns an object of class `betasandwich` which is a list with the following elements:

call Function call.

args Function arguments.

lm_process Processed `lm` object.

gamma_n Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

gamma_hc Asymptotic covariance matrix HC correction.

gamma Asymptotic covariance matrix of the sample covariance matrix.

acov Asymptotic covariance matrix of the standardized slopes.

vcov Sampling covariance matrix of the standardized slopes.

est Vector of standardized slopes.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
# Methods -----------------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

BetaN *Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality*

Description

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality

Usage

```
BetaN(object, alpha = c(0.05, 0.01, 0.001), std_y = TRUE, std_x = NULL)
```

Arguments

object	Object of class lm.
alpha	Numeric vector. Significance level α.
std_y	Logical. Standardize y. With std_y = FALSE, y is left in its original units ($b_j\sigma_{x_j}$).
std_x	Character vector or NULL. Names of the regressors to standardize. NULL (default) standardizes all regressors; character(0) standardizes none (b_j/σ_y). Leaving a dummy variable out keeps its 0-1 contrast. The delta method accounts for the sampling variability of the standard deviations.

Details

Note that while the calculation in BetaN() is different from betaDelta::BetaDelta() with type = "mvn", the results are numerically equivalent. BetaN() assumes multivariate normality. BetaHC() is recommended in most situations.

Value

Returns an object of class betasandwich which is a list with the following elements:

call Function call.

args Function arguments.

lm_process Processed lm object.

gamma_n Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

gamma_hc Asymptotic covariance matrix HC correction.

gamma Asymptotic covariance matrix of the sample covariance matrix.

acov Asymptotic covariance matrix of the standardized slopes.

vcov Sampling covariance matrix of the standardized slopes.

est Vector of standardized slopes.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaN(object)
# Methods -----------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

coef.betasandwich *Standardized Regression Slopes*

Description

Standardized Regression Slopes

Usage

```
## S3 method for class 'betasandwich'
coef(object, ...)
```

Arguments

`object`	Object of class `betasandwich`.
`...`	additional arguments.

Value

Returns a vector of standardized regression slopes.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
coef(std)
```

---

coef.essandwich *Estimates for an Object of Class* essandwich

---

Description

Estimates for an Object of Class essandwich

Usage

```
## S3 method for class 'essandwich'
coef(object, ...)
```

Arguments

object	Object of class essandwich.
...	additional arguments.

Value

Returns a vector of estimates.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
coef(out)
```

`confint.betasandwich` *Confidence Intervals for Standardized Regression Slopes*

Description

Confidence Intervals for Standardized Regression Slopes

Usage

```
## S3 method for class 'betasandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

Arguments

`object`	Object of class `betasandwich`.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.
`...`	additional arguments.

Value

Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
confint(std, level = 0.95)
```

confint.essandwich *Confidence Intervals for an Object of Class* essandwich

Description

Confidence Intervals for an Object of Class essandwich

Usage

```
## S3 method for class 'essandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

Arguments

object	Object of class essandwich.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
...	additional arguments.

Value

Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
confint(out, level = 0.95)
```

`DeltaRSqSandwich` *Estimate Improvement in R-Squared and the Corresponding Sampling Covariance Matrix*

Description

Estimate Improvement in R-Squared and the Corresponding Sampling Covariance Matrix

Usage

```
DeltaRSqSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

Arguments

`object`	Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions.
`alpha`	Numeric vector. Significance level α.

Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in `object` (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent). Two or more regressors are required.

Value

Returns an object of class `essandwich` which is a list with the following elements:

call Function call.

fit The argument `object`.

args Function arguments.

vcov Sampling covariance matrix of improvement in R-squared (squared semipartial correlation coefficients).

est Vector of improvement in R-squared (squared semipartial correlation coefficients).

fun Function used ("DeltaRSqSandwich").

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- DeltaRSqSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

`DiffBetaSandwich` *Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix*

Description

Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix

Usage

```
DiffBetaSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

Arguments

`object`	Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions.
`alpha`	Numeric vector. Significance level α.

Value

Returns an object of class `essandwich` (and `diffbetasandwich`) which is a list with the following elements:

call Function call.

fit The argument `object`.

args Function arguments.

vcov Sampling covariance matrix of differences of standardized slopes.

est Vector of differences of standardized slopes.

fun Function used ("DiffBetaSandwich").

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Beta Sandwich Functions: BetaADF(), BetaHC(), BetaN(), DeltaRSqSandwich(), FSqSandwich(), PCorSandwich(), RSqSandwich(), SCorSandwich()

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
# Methods ------------------------------------------------------
print(diff)
summary(diff)
coef(diff)
vcov(diff)
confint(diff, level = 0.95)
```

FSqSandwich *Estimate Cohen's f-Squared and Local f-Squared and the Corresponding Sampling Covariance Matrix*

Description

Estimate Cohen's f-Squared and Local f-Squared and the Corresponding Sampling Covariance Matrix

Usage

```
FSqSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

Arguments

object	Object of class betasandwich, that is, the output of the BetaHC(), BetaN(), or BetaADF() functions.
alpha	Numeric vector. Significance level α.

Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in object (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent).

Value

Returns an object of class essandwich which is a list with the following elements:

call Function call.

fit The argument object.

args Function arguments.

vcov Sampling covariance matrix of Cohen's f-squared (fsq) and the local f-squared of each regressor.

est Vector of Cohen's f-squared (fsq) and the local f-squared of each regressor.

fun Function used ("FSqSandwich").

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Beta Sandwich Functions: BetaADF(), BetaHC(), BetaN(), DeltaRSqSandwich(), DiffBetaSandwich(), PCorSandwich(), RSqSandwich(), SCorSandwich()

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- FSqSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

---

nas1982 *1982 National Academy of Sciences Doctoral Programs Data*

Description

1982 National Academy of Sciences Doctoral Programs Data

Usage

```
nas1982
```

Format

Ratings of 46 doctoral programs in psychology in the USA with the following variables:

QUALITY Program quality ratings.

NFACUL Number of faculty members in the program.

NGRADS Number of program graduates.

PCTSUPP Percentage of program graduates who received support.

PCTGRT Percent of faculty members holding research grants.

NARTIC Number of published articles attributed to program faculty member.

PCTPUB Percent of faculty with one or more published article.

References

National Research Council. (1982). *An assessment of research-doctorate programs in the United States: Social and behavioral sciences*. doi:10.17226/9781. Reproduced with permission from the National Academy of Sciences, Courtesy of the National Academies Press, Washington, D.C.

`PCorSandwich` *Estimate Squared Partial Correlation Coefficients and the Corresponding Sampling Covariance Matrix*

Description

Estimate Squared Partial Correlation Coefficients and the Corresponding Sampling Covariance Matrix

Usage

```
PCorSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

Arguments

`object`	Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions.
`alpha`	Numeric vector. Significance level α.

Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in `object` (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent). Two or more regressors are required.

Value

Returns an object of class `essandwich` which is a list with the following elements:

call Function call.

fit The argument `object`.

args Function arguments.

vcov Sampling covariance matrix of squared partial correlation coefficients.

est Vector of squared partial correlation coefficients.

fun Function used ("PCorSandwich").

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `RSqSandwich()`, `SCorSandwich()`

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- PCorSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

`print.betasandwich` *Print Method for an Object of Class* `betasandwich`

Description

Print Method for an Object of Class `betasandwich`

Usage

```
## S3 method for class 'betasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

Arguments

x	Object of class betasandwich.
alpha	Numeric vector. Significance level α. If alpha = NULL, use the argument alpha used in x.
digits	Digits to print.
...	additional arguments.

Value

Prints a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
print(std)
```

---

print.essandwich *Print Method for an Object of Class* essandwich

---

Description

Print Method for an Object of Class essandwich

Usage

```
## S3 method for class 'essandwich'
print(x, alpha = NULL, digits = 4, ...)
```

Arguments

x	Object of class essandwich.
alpha	Numeric vector. Significance level α. If alpha = NULL, use the argument alpha used in x.
digits	Digits to print.
...	additional arguments.

Value

Prints a matrix of estimates, standard errors, test statistics, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
print(out)
```

---

print.summary.essandwich

Print Method for an Object of Class `summary.essandwich`

Description

Print Method for an Object of Class `summary.essandwich`

Usage

```
## S3 method for class 'summary.essandwich'
print(x, ...)
```

Arguments

`x`	Object of class `summary.essandwich`.
`...`	additional arguments.

Value

Prints a matrix of estimates, standard errors, test statistics, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
summary(out)
```

RSqSandwich *Estimate R-Squared and Adjusted R-Squared and the Corresponding Sampling Covariance Matrix*

Description

Estimate R-Squared and Adjusted R-Squared and the Corresponding Sampling Covariance Matrix

Usage

```
RSqSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

Arguments

object	Object of class betasandwich, that is, the output of the BetaHC(), BetaN(), or BetaADF() functions.
alpha	Numeric vector. Significance level α.

Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in object (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent).

Value

Returns an object of class essandwich which is a list with the following elements:

call Function call.

fit The argument object.

args Function arguments.

vcov Sampling covariance matrix of R-squared and adjusted R-squared.

est Vector of R-squared and adjusted R-squared.

fun Function used ("RSqSandwich").

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Beta Sandwich Functions: BetaADF(), BetaHC(), BetaN(), DeltaRSqSandwich(), DiffBetaSandwich(), FSqSandwich(), PCorSandwich(), SCorSandwich()

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

`SCorSandwich` *Estimate Semipartial Correlation Coefficients and the Corresponding Sampling Covariance Matrix*

Description

Estimate Semipartial Correlation Coefficients and the Corresponding Sampling Covariance Matrix

Usage

```
SCorSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

Arguments

`object`	Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions.
`alpha`	Numeric vector. Significance level α.

Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in `object` (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent). Two or more regressors are required.

Value

Returns an object of class `essandwich` which is a list with the following elements:

call Function call.

fit The argument `object`.

args Function arguments.

vcov Sampling covariance matrix of semipartial correlation coefficients.

est Vector of semipartial correlation coefficients.

fun Function used ("SCorSandwich").

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- SCorSandwich(std)
# Methods -----------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

`summary.betasandwich` *Summary Method for an Object of Class* `betasandwich`

Description

Summary Method for an Object of Class `betasandwich`

Usage

```
## S3 method for class 'betasandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

Arguments

`object`	Object of class `betasandwich`.
`alpha`	Numeric vector. Significance level α. If `alpha = NULL`, use the argument `alpha` used in `object`.
`digits`	Digits to print.
`...`	additional arguments.

Value

Returns a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
summary(std)
```

---

summary.essandwich *Summary Method for an Object of Class* essandwich

---

Description

Summary Method for an Object of Class essandwich

Usage

```
## S3 method for class 'essandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

Arguments

object	Object of class essandwich.
alpha	Numeric vector. Significance level α. If alpha = NULL, use the argument alpha used in object.
digits	Digits to print.
...	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
summary(out)
```

vcov.betasandwich *Sampling Covariance Matrix of the Standardized Regression Slopes*

Description

Sampling Covariance Matrix of the Standardized Regression Slopes

Usage

```
## S3 method for class 'betasandwich'
vcov(object, ...)
```

Arguments

object	Object of class betasandwich.
...	additional arguments.

Value

Returns a matrix of the variance-covariance matrix of standardized slopes.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
vcov(std)
```

vcov.essandwich *Sampling Covariance Matrix for an Object of Class* essandwich

Description

Sampling Covariance Matrix for an Object of Class essandwich

Usage

```
## S3 method for class 'essandwich'
vcov(object, ...)
```

Arguments

object	Object of class essandwich.
...	additional arguments.

Value

Returns a matrix of the variance-covariance matrix of the estimates.

Author(s)

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Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
vcov(out)
```

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